DOTE 6751 · GROUP PROJECT

SubSplit

Escrow-protected subscription sharing

A family plan is priced for six people. SubSplit makes five strangers safe enough to buy the other seats — money in escrow, released one month at a time, with a record both sides can point at.

11	15	18
relations	foreign keys	queries

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A household price for people who have no household

Six seats for the price of one and a half. The discount assumes a trust that does not exist yet.

THE MEMBER'S RISK

Paid for a year, holds nothing

The host owns the account. One password change, one silent removal, and twelve months of money sit with a stranger who has no obligation to answer.

THE HOST'S CHORE

Owes the platform, not the rent

Five people, one small transfer each, every month. Chasing friends for HK$18 costs more in awkwardness than the money is worth.

So the design question is not how to split a bill.

It is how to make a stranger's promise enforceable — with a database, not with goodwill.

How a group works, and where the money sits

Six seats, twelve prepaid months, held in escrow and released one month at a time.

1. **HOST VERIFIES THE BILL**
   The platform checks the official invoice before a group goes live.
2. **GROUP OPENS SIX SEATS**
   The host sets the price per seat per month; the group is listed as recruiting.
3. **MEMBERS PREPAY 12 MONTHS**
   Money lands in the escrow pool — not in the host's account.
4. **PLATFORM RELEASES 1/12**
   One month is paid to the host each cycle; the platform keeps a 5% fee.

ESCROW POOL · ALL GROUPS

HK$26,113

still held for members across 300 seats



58% already released to hosts

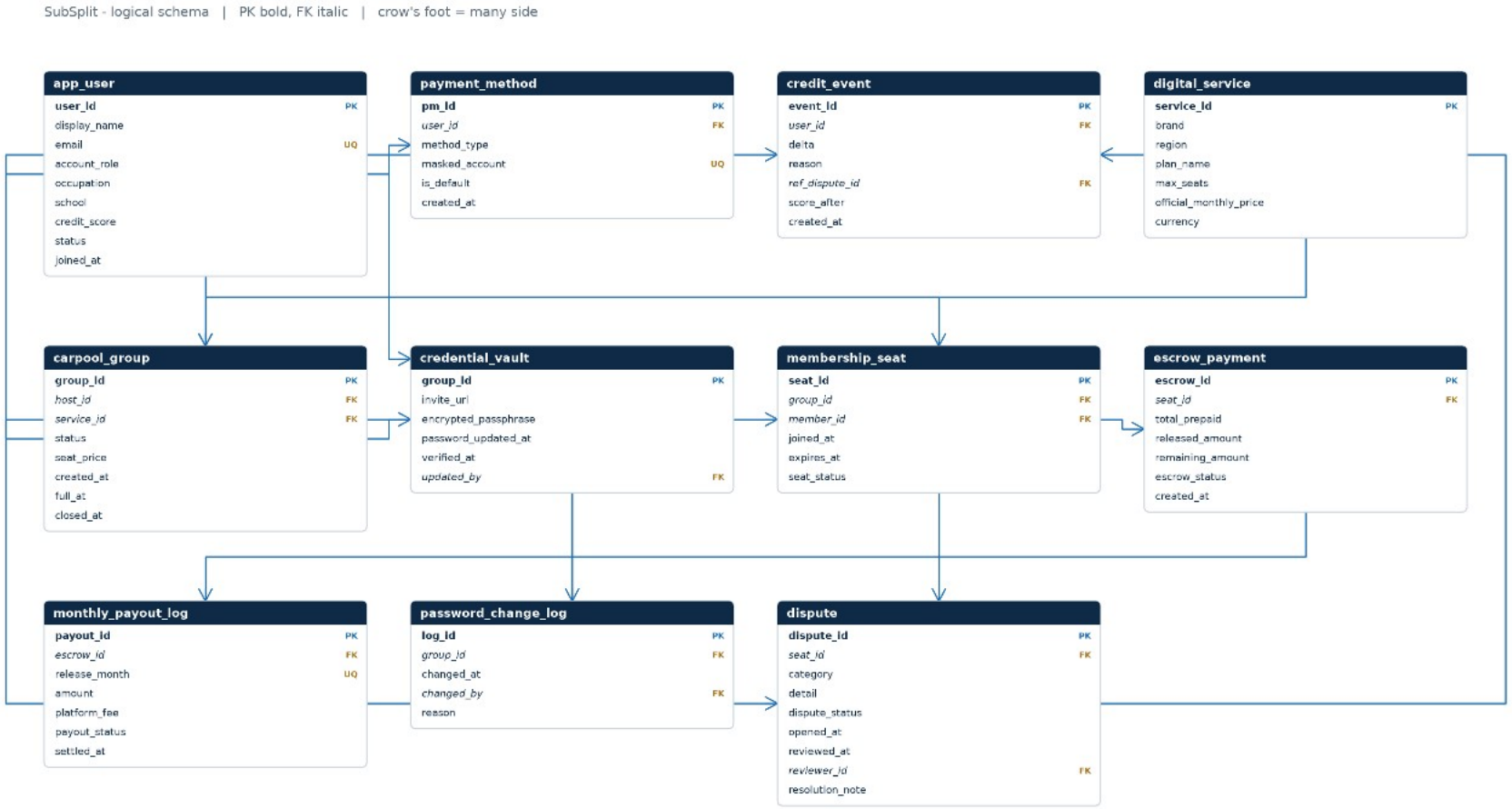
Prepaid by members	**HK$61,577**
Released to hosts	**HK$35,464**
Platform fee (5%)	**HK$1,772**

Frozen the moment a dispute is filed.

6	12	5%	1/12
SEATS PER GROUP	MONTHS PREPAID	PLATFORM FEE	RELEASED MONTHLY

Eleven relations, fifteen foreign keys

The whole schema at once. Module views follow, because eleven boxes on one page is a map, not a diagram.



WEAK ENTITY

credential_vault borrows group_id as its own key — no group, no credential.

ASSOCIATIVE ENTITY

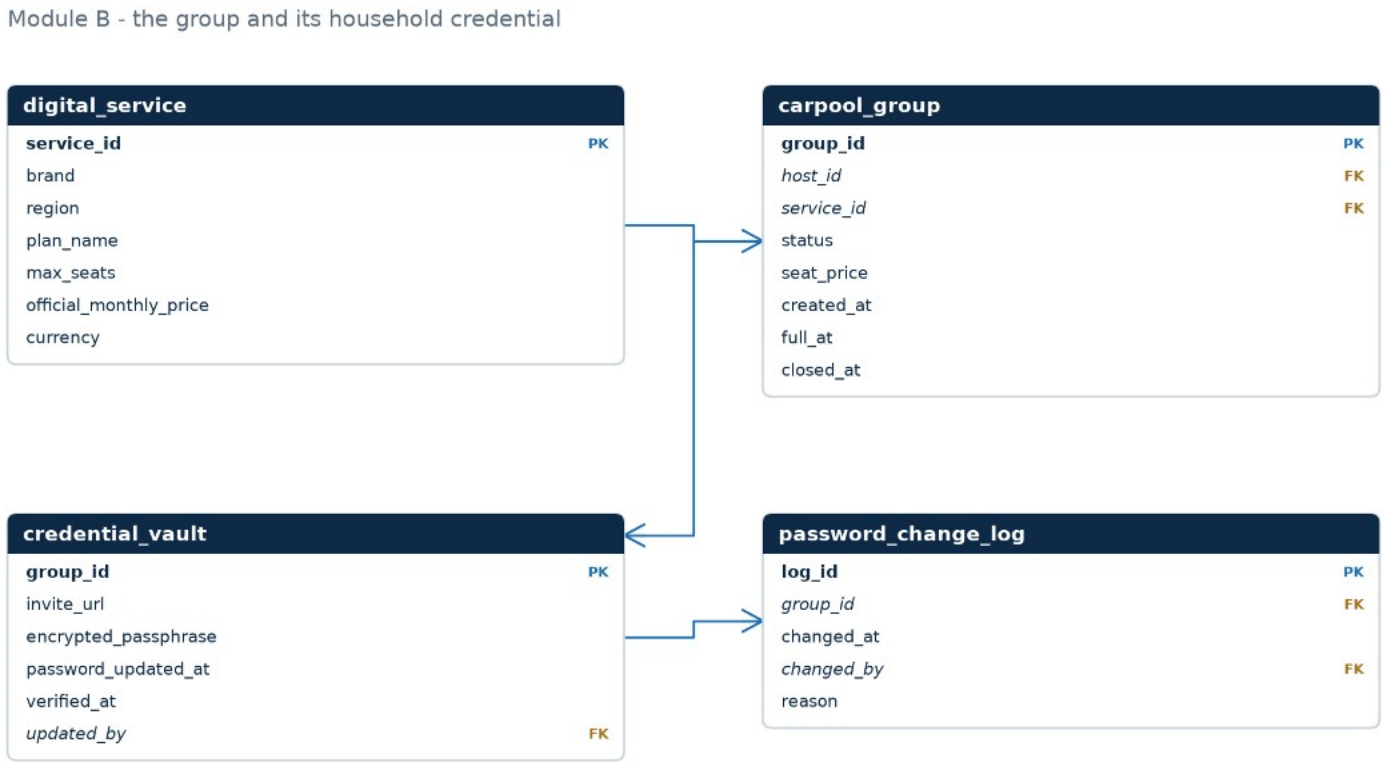
membership_seat is the M:N link promoted to an entity with its own life cycle.

LEDGER

monthly_payout_log keys on (escrow_id, release_month): a month cannot be paid twice.

The group and its credential

One host, one product, one credential: the 1:1 that makes a shared household auditable.



1:1 BY CONSTRUCTION

group_id is both the primary key and the foreign key of credential_vault — a second vault for the same group is impossible.

CAPACITY IS A PRODUCT FACT

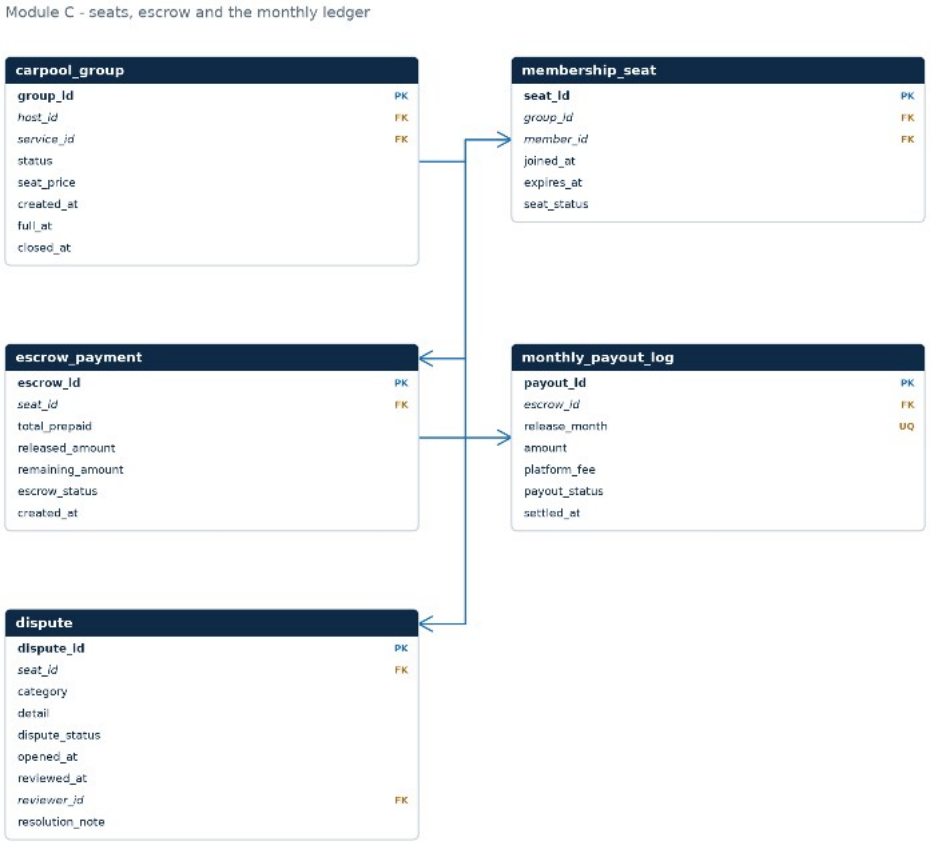
max_seats lives in digital_service, never copied into the group; the trigger joins to read it.

EVERY CHANGE IS JOURNALLED

trg_password_log writes a row whenever password_updated_at moves.

Seats, escrow and the monthly ledger

Money has three homes: what was prepaid, what has been released, and what is still owed.



1:1 SEAT → ESCROW

seat_id is unique in escrow_payment, so a seat has exactly one pot of money.

BALANCE IS A CACHE

remaining = total − released is a CHECK, not an assumption; Q11 audits every row.

COMPLAINT FREEZES MONEY

a dispute never deletes a payout; it changes escrow_status and blocks the next release.

What each table is for

Eleven tables, 76 attributes. Every one of them answers a question the business actually asks.

Table	Rows	Why it exists
app_user	300	identity, role, credit score
payment_method	379	several instruments, one default
digital_service	8	product dictionary: brand, region, seat ceiling
carpool_group	60	one host reselling one product
credential_vault	60	the household credential (weak entity)
membership_seat	300	who sits where, for how long
escrow_payment	300	the prepayment held for one seat
monthly_payout_log	2,172	every release, with its fee
password_change_log	74	credential history, for risk
dispute	33	complaints and their state machine
credit_event	109	why the score moved

READING THE TABLE

Three of the eleven are history tables — they exist because a value is not enough, the sequence matters:

· password_change_log answers “how many times in 30 days”, which a single timestamp cannot.

· monthly_payout_log answers “how much has actually been paid”, and makes double payment impossible.

· credit_event answers “why is this host at 60”, which a number alone cannot.

Eighteen queries, six on stage

Three families: lookups a member runs, analysis the business runs, and controls that only make sense over time.

6 **LOOKUPS**
find a group, a seat, a credential

7 **BUSINESS**
revenue, fill time, refunds, ranking

5 **ADVANCED**
CTEs, windows, anti-joins, audits

Q01 · PRECISE SEAT HUNTING

Recruiting groups on Spotify Hong Kong, exactly one seat left, host with a perfect score.

```
SELECT g.group_id, ds.brand, g.seat_price, u.credit_score,
       COUNT(s.seat_id) AS seats_taken,
       ds.max_seats - COUNT(s.seat_id) AS seats_left
FROM carpool_group g
JOIN digital_service ds USING (service_id)
JOIN app_user u ON u.user_id = g.host_id
LEFT JOIN membership_seat s
  ON s.group_id = g.group_id AND s.seat_status = 'ACTIVE'
WHERE g.status = 'RECRUITING' AND ds.brand = 'Spotify'
  AND ds.region = 'HK' AND u.credit_score = 100
GROUP BY g.group_id, ds.brand, g.seat_price, u.credit_score, ds.max_seats
HAVING seats_left = 1;
```

group	price	host	seats
1	20.22	Olivia CHU	5/6
46	21.63	Peter ZHANG	5/6

The search is narrow by design: a member with a year of money at stake wants the one group that is about to fill and whose host has never defaulted.

Fraud is a pattern, not a payment

Q05 joins credential history to complaints and returns the queue an administrator should open first.

group	host	service	status	score	pwd / 30d	complaints
26	Lok TSANG	Disney+	ACTIVE	60	6	3
34	Jason MA	Netflix	ACTIVE	39	6	2
12	Mandy FUNG	Disney+	FULL	60	5	3
9	Mei MA	Spotify	RECRUITING	60	5	3
8	Grace CHAN	Netflix	ACTIVE	60	4	6
35	Yitong CHEUNG	Apple One	FULL	30	3	4

READ IT AS A RULE, NOT A SCORE

Three credential changes in thirty days is what a host does when a paying member has been locked out. Two “cannot log in” complaints on the same group is what the members do next.

The query deliberately does not refund, suspend or punish: it produces a ranked queue. The decision stays with an administrator, and the state machine in dispute makes that decision visible.

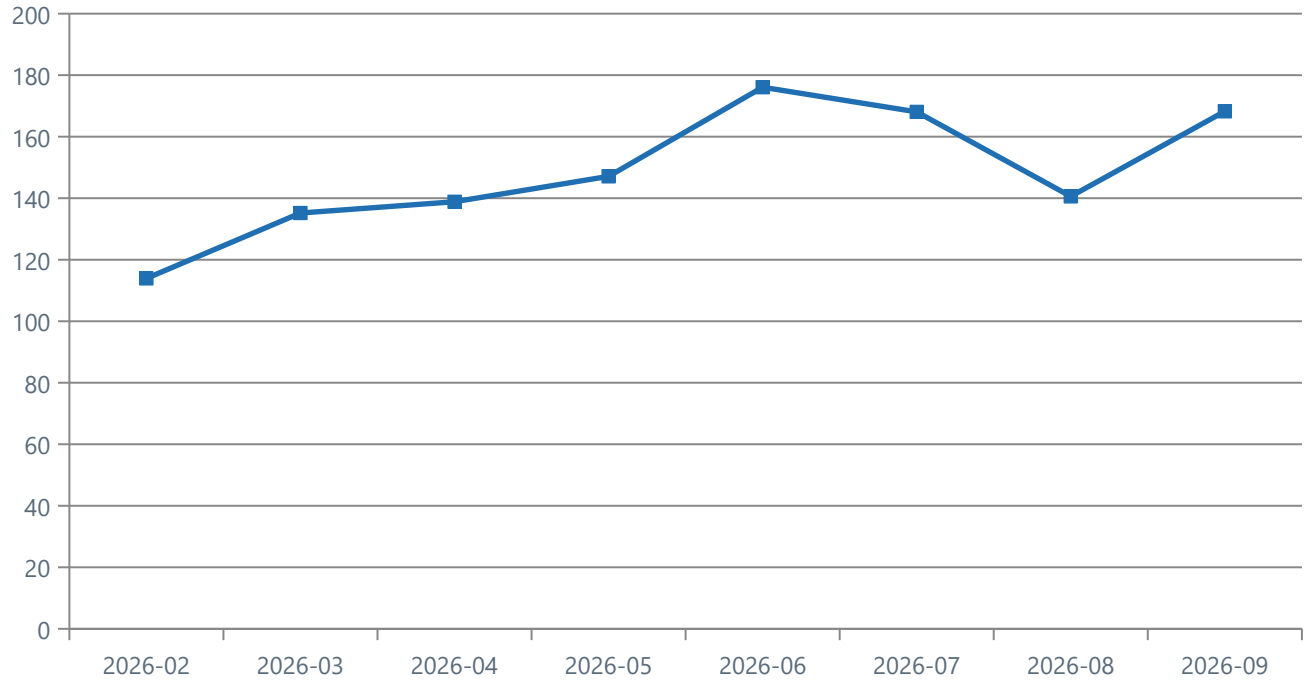
33
DISPUTES ON FILE

6
GROUPS FLAGGED

0
AUTOMATIC REFUNDS

What the platform earns, and what people actually buy

Q04 aggregates the ledger by month; Q02 ranks services by the seats students occupy.



Peak month: HK$176 in fees. The August dip is a month with fewer releases, not a decline in the business.

service	seats	groups
Netflix HK	42	11
ChatGPT HK	26	9
YouTube HK	24	9
Spotify US	22	7
Spotify HK	20	6

Netflix-HK leads because it has the widest price gap between the individual plan and a six-seat family plan — the same gap that motivates the platform.

Two questions that only make sense over time

Both need a window function: one looks at the previous month, the other at the previous row in the same escrow.

Q10 · CONSECUTIVE BLOCKED MONTHS

LAG(payout_status) over each escrow finds groups that stopped paying twice in a row — a stronger signal than a single blocked month.

```
WITH ledger AS (
  SELECT escrow_id, release_month, payout_status,
         LAG(payout_status) OVER (
           PARTITION BY escrow_id
           ORDER BY release_month) AS prev
  FROM monthly_payout_log)
SELECT escrow_id, release_month FROM ledger
WHERE payout_status = 'BLOCKED'
  AND prev = 'BLOCKED';
```

26 rows across the seeded data — every one of them belongs to a group already under dispute.

Q17 · MONTH-OVER-MONTH FEE GROWTH

LAG over an aggregated month series: the fee this month compared with the fee last month, in one pass.

month	fee	prev	Δ %
2026-04	138.86	135.19	2.7
2026-05	147.11	138.86	5.9
2026-06	176.09	147.11	19.7
2026-07	168.06	176.09	-4.6
2026-08	140.59	168.06	-16.3
2026-09	168.24	140.59	19.7

Both are plain SELECTs: nothing is precomputed, nothing is cached.

The same function answers two different questions — that is the point of learning it once, not twice.

What this design does not do

Honest scope, and the three places we would go next.

MODELLED, NOT IMPLEMENTED

— Escrow as a state machine, not a payment gateway

— Credential as encrypted text, not a secrets manager

— Host bill verification as a date, not a process

KNOWN LIMITS

— No partial months: leave on day 20, keep the month

— No cross-region pricing: comparisons stay within a region

— A determined host can still lie — we only make it visible

NEXT

— Materialised view for the market listing

— Partition the ledger by year once it passes ~10M rows

— Archival policy for CLOSED groups older than two years

Every limit above is a decision we made on purpose, and each has a next step — which is why the design is finished, not unfinished.

Task allocation

Seven members, one schema. Hours are estimates agreed at the start; difficulty is 1–5.

Student	ID	Task	Difficulty	Hours
Xuanmo AN	1155260677	Conceptual E-R model: entities, relationships, cardinality limits, weak-entity decision	4	11
Yi HOU	1155261383	E-R to relations: §6.7 mapping rules, key placement, 3NF/BCNF proof	5	13
Zhao Han HUANG	1155272384	DDL and integrity constraints: CHECK, unique keys, referential actions, indexes	5	12
Jinyu LI	1155261567	Triggers and views: seat ceiling, default instrument, credential journal, balance upkeep	5	12
Yitong NI	1155265810	Deterministic seed generator and the transaction episodes	4	10
Shujing YE	1155261750	Eighteen queries: grouping, window functions, CTEs, result capture	4	11
Shiyao ZHANG	1155261823	Acceptance suite, fault injection and ledger reconciliation	4	10

Total 79 hours. Difficulty follows the number of concepts a task touches, not the number of lines written.

If you ask us this, here is the answer

Prepared responses to the seven questions this design invites.

Why store the balance if it is derivable?

It is read on every dashboard view; the ledger is written monthly. A CHECK makes the identity enforceable, and Q11 audits it.

Why manual review instead of auto-refund?

Two coordinated friends could drain any host. Detection is automatable; judgement is not.

Is this 3NF?

BCNF everywhere. The proof lists every functional dependency and every candidate key.

Why not NoSQL?

Money, seats and state transitions need multi-row invariants; that is the relational model's home ground.

How do you stop double-selling a seat?

UNIQUE(group_id, member_id) plus a trigger that counts seats against digital_service.max_seats.

What happens if a host deletes the account?

RESTRICT on carpool_group.host_id: the group must be closed and its escrow settled first.

Where is the evidence it runs?

subsplit_db on MySQL 8.0.40; every query executed, results saved in query_results/.